



Paper ID : 261195

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Subject Code: BCE401

Roll No:

BTECH

(SEM. IV) THEORY EXAMINATION 2025-26
MATERIALS, TESTING & CONSTRUCTION PRACTICES

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief.

02 x 7 = 14

Q no.	Question	CO	Level
a.	Define workability of concrete. List factors affecting it.	1	K2
b.	What are the properties of good building stone?	1	K2
c.	Define hydration of cement.	1	K1
d.	What is seasoning of timber? Why is it necessary?	1	K2
e.	List types of paints used in construction.	2	K1
f.	Define safe bearing capacity of soil.	3	K1
g.	What is damp proofing?	5	K1

SECTION B

2. Attempt any *three* of the following:

07 x 3 = 21

Q no.	Question	CO	Level
a.	Explain manufacturing process of bricks with neat sketch.	1	K2
b.	Discuss properties and uses of fly ash and GGBS in concrete.	2	K3
c.	Explain types of foundations with suitability.	3	K2
d.	Describe different types of flooring materials and their applications.	4	K2
e.	Explain types of roofs with advantages and disadvantages.	4	K3

SECTION C

3. Attempt any *one* part of the following:

07 x 1 = 07

Q no.	Question	CO	Level
a.	Explain cement manufacturing process (dry process) with flow diagram.	1	K3
b.	Discuss grading of aggregates and its importance in concrete.	1	K3

4. Attempt any *one* part of the following:

07 x 1 = 07

Q no.	Question	CO	Level
a.	Explain principles of building planning orientation, ventilation, lighting	3	K4
b.	Describe different types of masonry bonds with sketches.	3	K2

5. Attempt any *one* part of the following:

07 x 1 = 07

Q no.	Question	CO	Level
a.	A concrete mix has ratio 1:2:4. Calculate quantity of cement required for 1 m ³ concrete. Assume dry volume = 1.54 × wet volume.	1	K3
b.	A wall of size 5 m × 3 m × 0.3 m is to be constructed. Calculate	3	K3



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	number of bricks required (size of brick = 19×9×9 cm, mortar = 10%).		
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6. Attempt any *one* part of the following:**07 x 1 = 07**

Q no.	Question	CO	Level
a.	Explain plastering process and types of plaster finishes.	5	K2
b.	Describe plumbing system of a building with neat sketch.	5	K3

7. Attempt any *one* part of the following:**07 x 1 = 07**

Q no.	Question	CO	Level
a.	Explain different types of foundations (shallow and deep) with neat sketches and their suitability.	3	K2
b.	Describe types of walls load bearing, partition and cavity walls with their functions and advantages.	4	K2